



CONTACT

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EDUCATION

2017 - 2025

Universiteit van Amsterdam

- Research MSc Brain & Cognitive Sciences (Cum Laude)
- BSc Cognition, Language and Communication
- Minor Artificial Intelligence

PUBLICATIONS

Fresen, A. J., Zuidema, W., Heilbron, M., & de Heer Kloots, M. (2024). *Language models that accurately represent syntactic information align better with brain activity*. Proceedings of the 46th Annual Meeting of the Cognitive Science Society.

SKILLS

AI Architecture and Systems

- AI system design and implementation
- Data infrastructure · RAG applications

Technical

- Python · R · PyTorch · Tensorflow · API integrations · High Performance Computing

Analysis and Statistics

- Regression · LM probing · Concept erasure · Representational Similarity Analysis · Searchlight analysis · ROI analysis · EEG analysis

Soft skills

- Presenting · Interdisciplinary collaboration · Supervising students

LANGUAGES

- Dutch (fluent)
- English (fluent)
- French (basic)

BRAM FRESSEN

COMPUTATIONAL LINGUIST & DATA ANALYST

PROFILE

Interdisciplinary researcher with a special fondness for statistics, artificial intelligence, and linguistics.

EXPERIENCE

ClickValue - Consultant | Data- and AI specialist 2024 - PRESENT

- Designed the architecture and implementation of an AI agent using retrieval augmented generation, custom Python and JavaScript tools and workflow orchestration in N8N.
- Developed a causal impact pipeline in R to detect effects of website changes on key metrics.
- Developed and deployed multiple web apps and pipelines.
- Designed and analyzed A/B experiments, translated statistical results into clear insights and delivered recommendations to clients.

University of Amsterdam (UvA) - Research intern 2022 - 2025 cognitive/neuro-AI & explainability

Institute for Logic, Language and Computation (ILLC)

- Designed and implemented an NLP pipeline for research on model-brain alignment between LLMs and fMRI data under supervision of Dr. Jelle Zuidema & MSc Marianne de Heer Kloots.
- Trained and evaluated an advanced dependency probe (DepProbe).
- Results published and presented at an international conference (CogSci 2024).

Amsterdam Brain and Cognition (ABC Labs)

- Built an NLP pipeline for causal probing of LLM representations in relation to fMRI data in Python under supervision of Dr. Micha Heilbron.
- Applied concept erasure (Residual removal and LEACE) to analyze and manipulate conceptual representations.
- Applied various probes to evaluate different language capabilities.

University of Amsterdam (UvA) - Teaching assistant 2018 - 2024

BSc

- Statistics in R (2x), Human Cognition, Advanced Statistics, Knowledge Application, Human Cognition, and Research Project 1.

MSc

- Research Design and Statistics

University of Amsterdam (UvA) - Research intern 2022 psycholinguistics

- Psycholinguistic research under supervision of Dr. Monique Flecken.
- Responsible for hypothesis formulation, EEG data collection, and data analysis.